HMGP 7622

Rethinking data analysis

A researcher's guide to avoiding missteps and misuse

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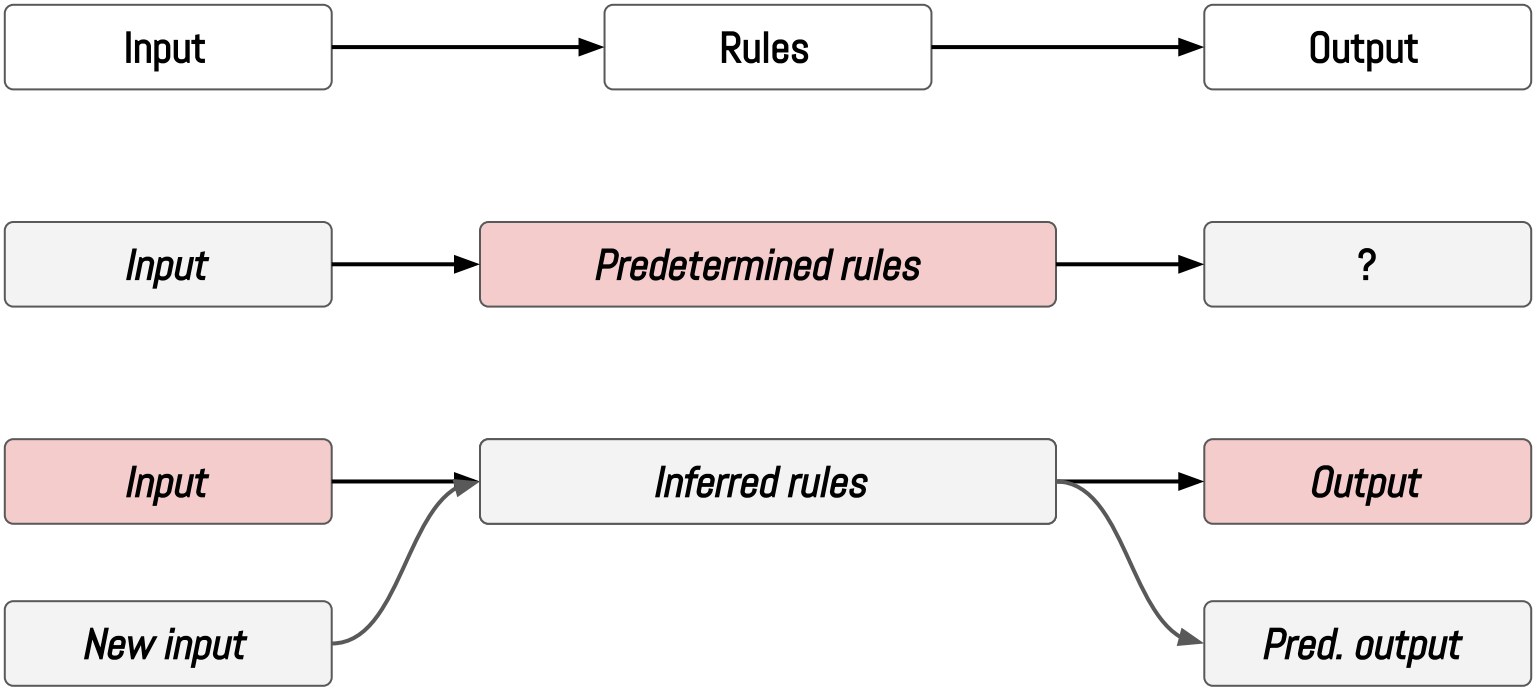
Day 09

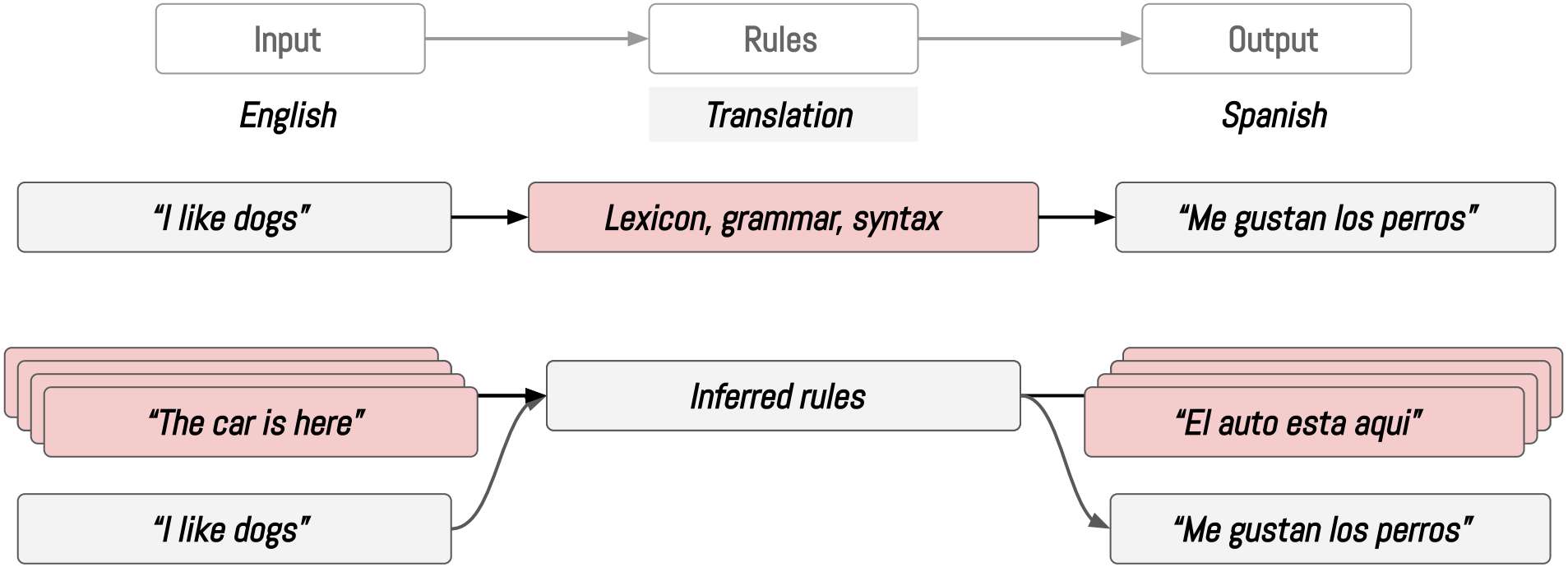
High-dimensional data & ML pitfalls

- What is ML?
- Curse of dimensionality
- Overfitting and model selection
- Common ML pitfalls
- Overall mitigation strategies

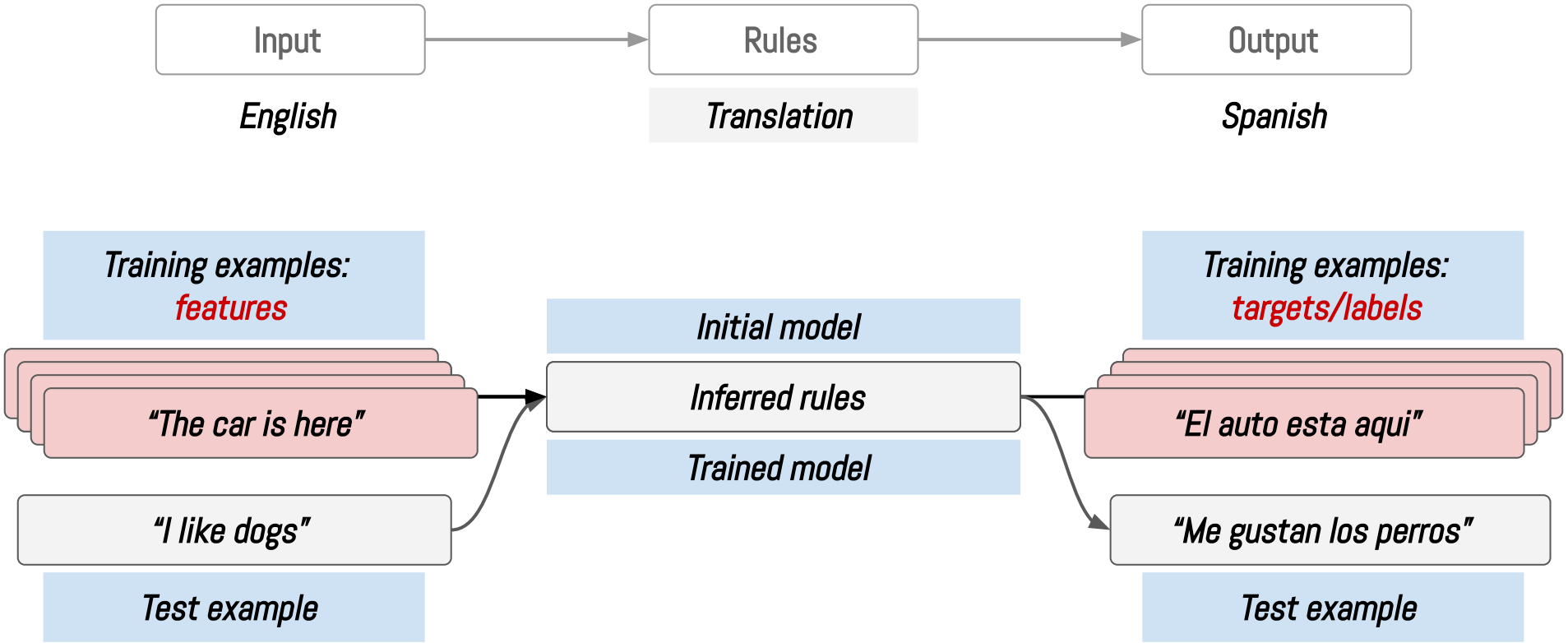
What is ML?


Input
Rules
Output
Input
Predetermined rules
?
Input
Inferred rules
Output
New input
Pred. output

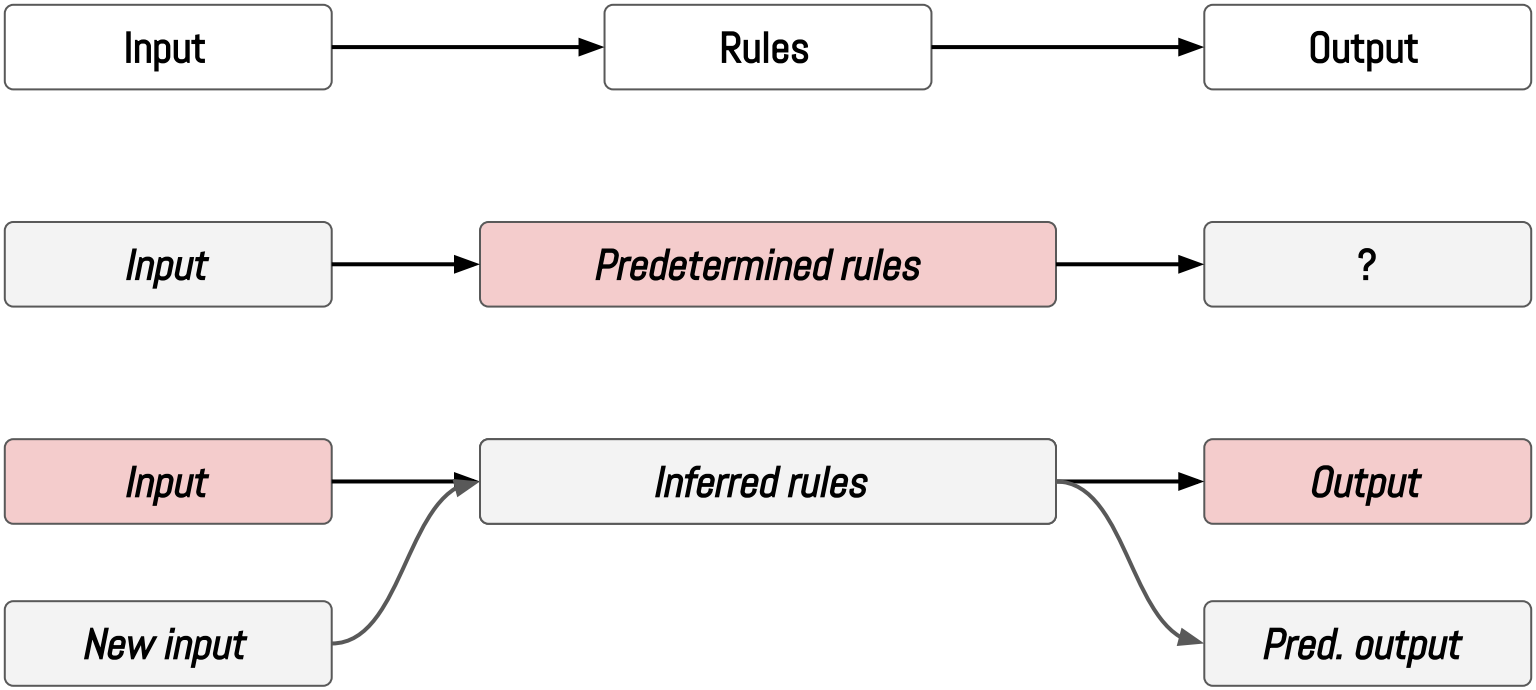
What is ML?


Input
Rules
Output
English
Translation
Spanish
"I like dogs"
Lexicon, grammar, syntax
"Me gustan los perros"
"The car is here"
"I like dogs"
Inferred rules
"El auto esta aqui"
"Me gustan los perros"

What is ML?

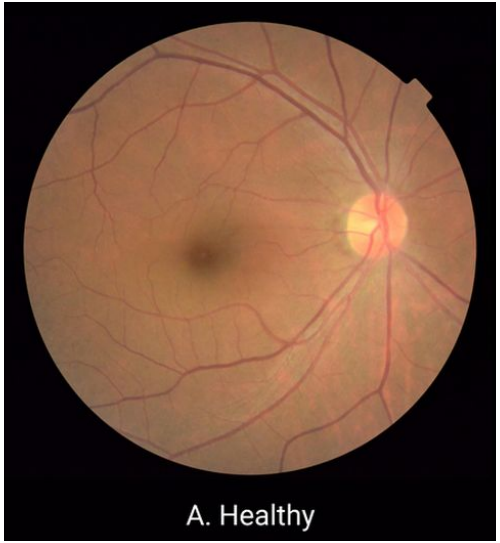

Input
Rules
Output
English
Translation
Spanish
Training examples:
features
"The car is here"
"I like dogs"
Test example
Initial model
Inferred rules
Trained model
Training examples:
targets/labels
"El auto esta aqui"
"Me gustan los perros"
Test example

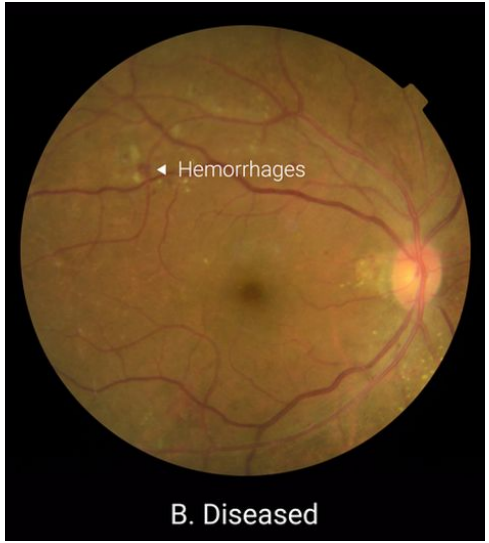
What is ML?



Exercise : connect this paradigm to fitting a regression line, given X & Y values. What are 'input', 'output', & 'inferred rules'?

Predicting diabetic retinopathy





Diagnosed by a **highly trained doctor** examining a retinal scan of the eye.

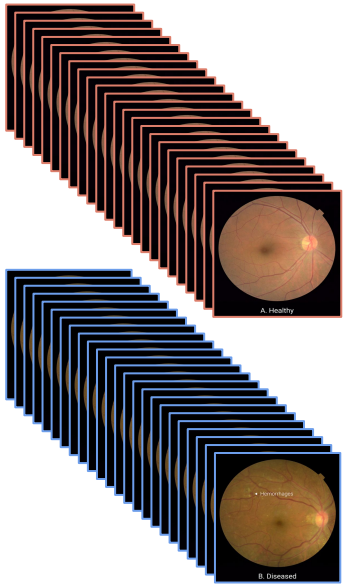
If caught early ⟶ effective treatments

If undetected ⟶ progresses into irreversible blindness

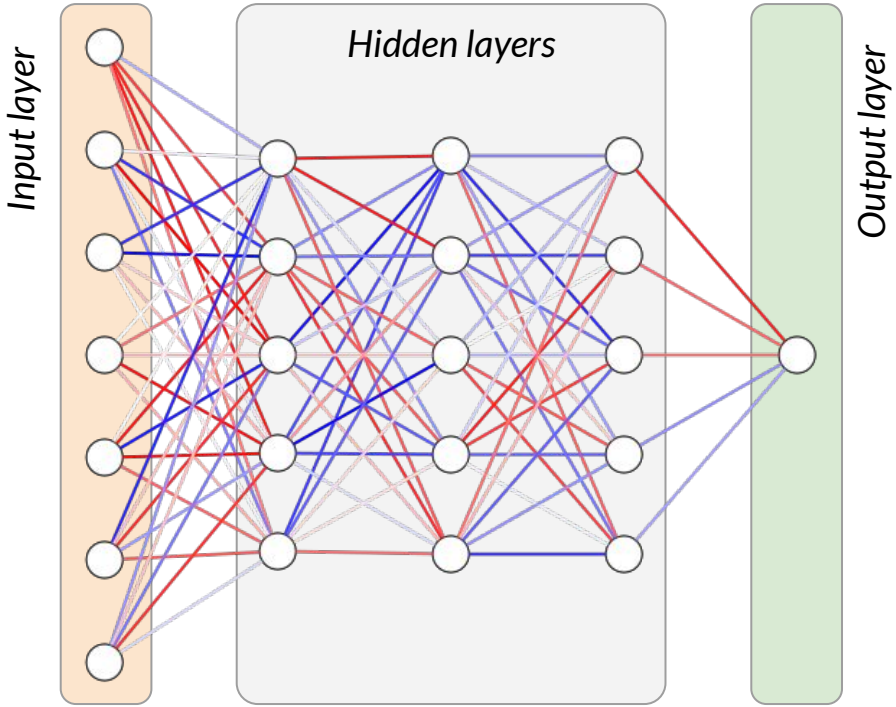
Not enough doctors available.

Predicting diabetic retinopathy

Data: *Thousands* of retina images



Machine learning **algorithm**

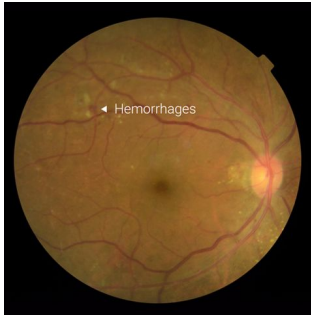


Labels

"**Healthy**" or "**Diseased**"

Predicting diabetic retinopathy

Data: A *new* patient's retina image



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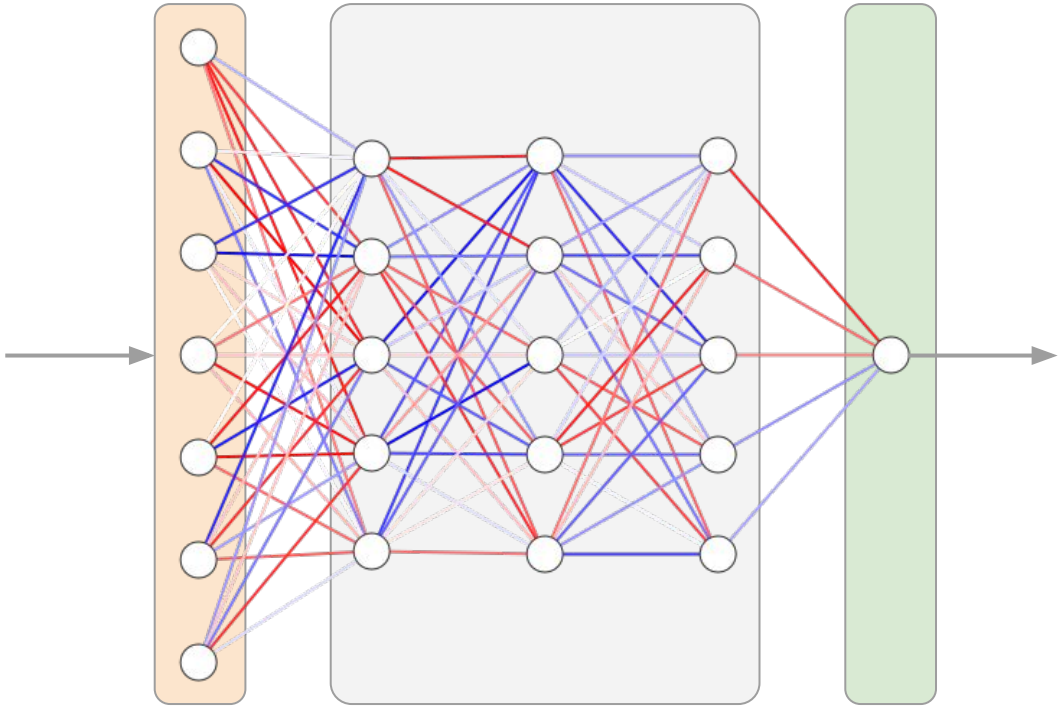
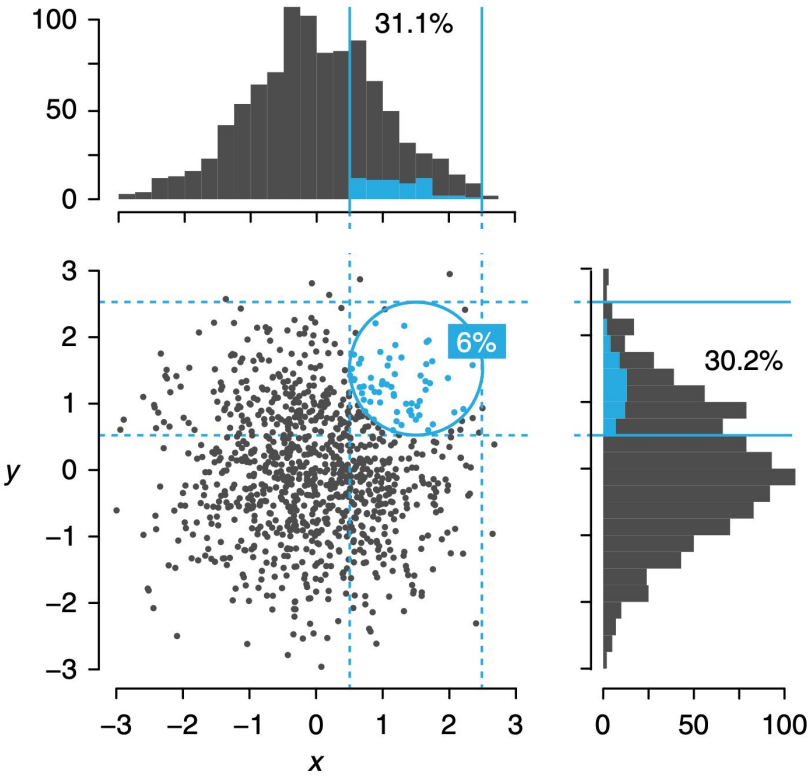


Prediction

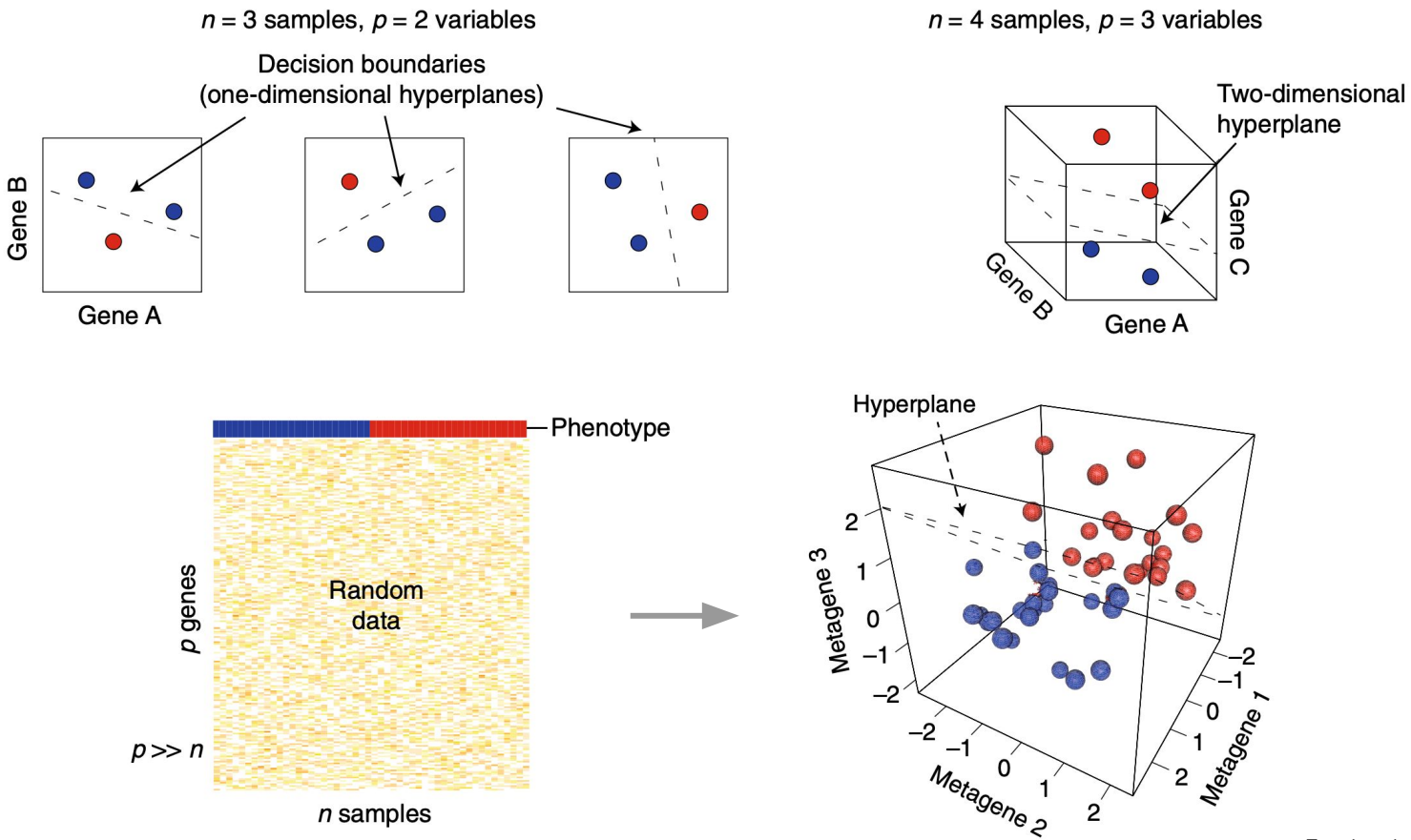
"**Diseased** " with 89% probability

Data tend to be sparse in high dimensions



Altman & Krzywinski (2018) *Nat. Meth.*

The curse of dimensionality

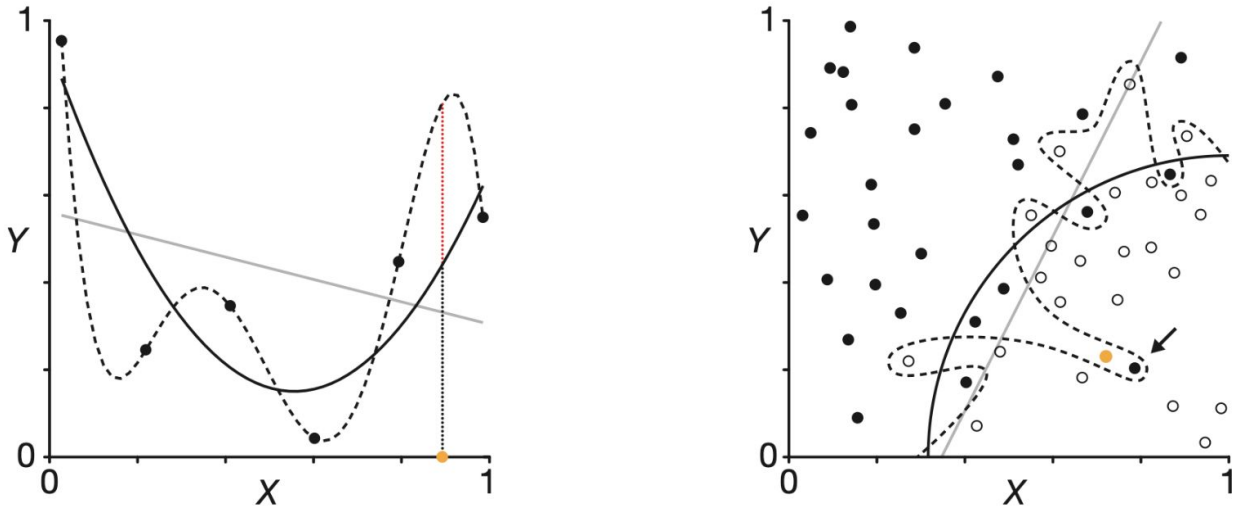


Teschendorff (2019) *Nat. Mat.*

Underfitting vs overfitting

Underfitting : model that is too simple to capture the underlying relationships

Overfitting : model that is too sensitivity to noise in the data & captures random fluctuations



As a model becomes more complex, underfitting generally decreases while overfitting increases.

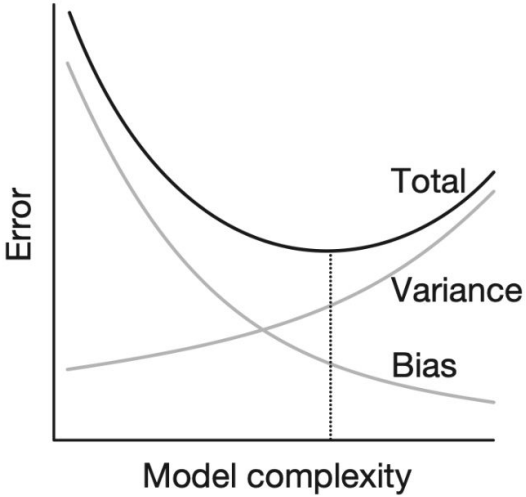
Lever et al., (2016) *Nat. Meth.*

The bias-variance tradeoff

Bias : the error introduced by a predictive model that is too simple to capture the underlying relationships within a system.

Variance : the error caused by a model's extreme sensitivity to noise in the data, thus capturing random fluctuations rather than true pattern.

The bias-variance tradeoff : as a model becomes more complex, bias generally decreases while variance increases.



Goal of model selection : find an appropriate level of complexity that balances bias and variance to minimize the total predictive error.

Lever et al., (2016) *Nat. Meth.*

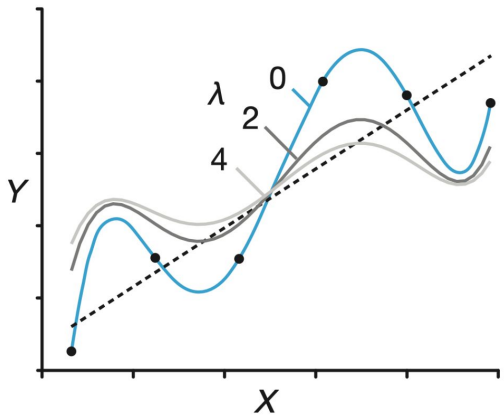
The bias-variance tradeoff

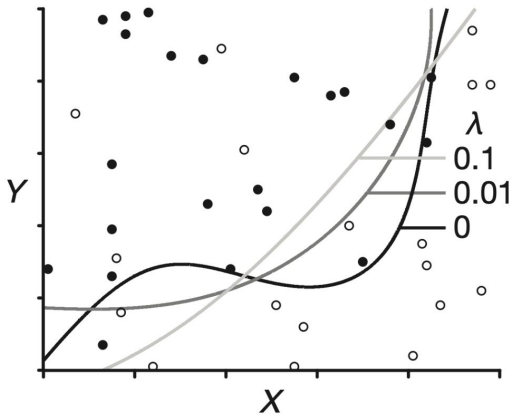
With high-dimensional data, models often have too much "flexibility" → easy to overfit!

Model's objective function ≈ minimize (Error + Complexity)

Regularization : add a penalty term to the model's objective function → a "limited budget" for the model to spend on its parameters → limiting model complexity (magnitude & number)

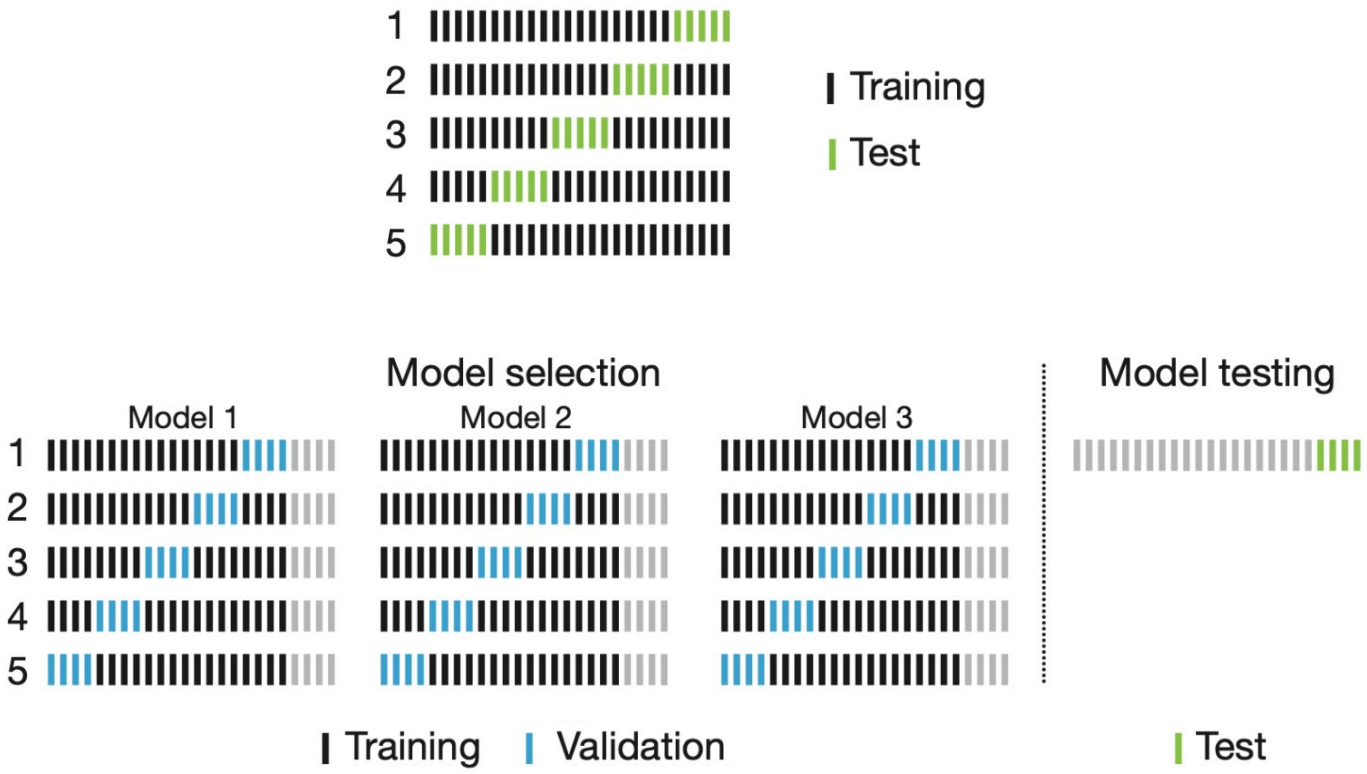
λ: penalty parameter, regularization parameter, or complexity parameter





Lever et al., (2016) *Nat. Meth.*

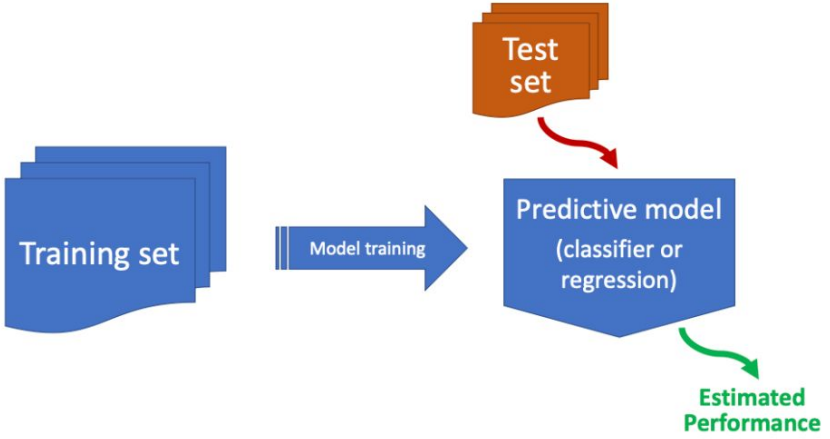
Cross-validation & model selection

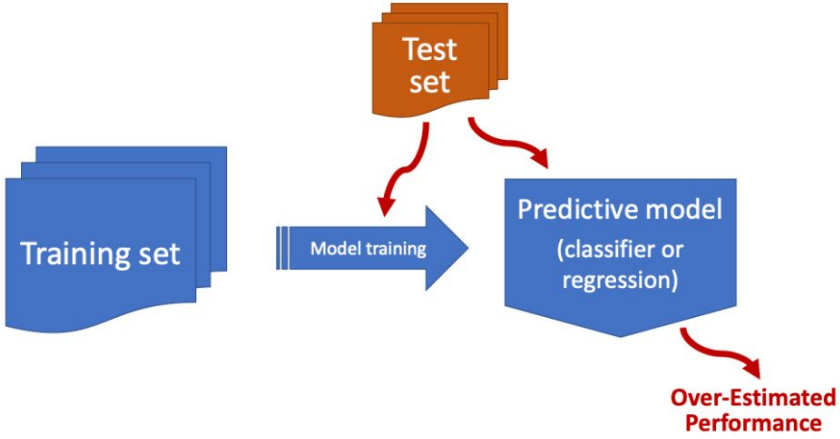


Lever et al., (2016) *Nat. Meth.*

Common ML pitfalls

Data leakage

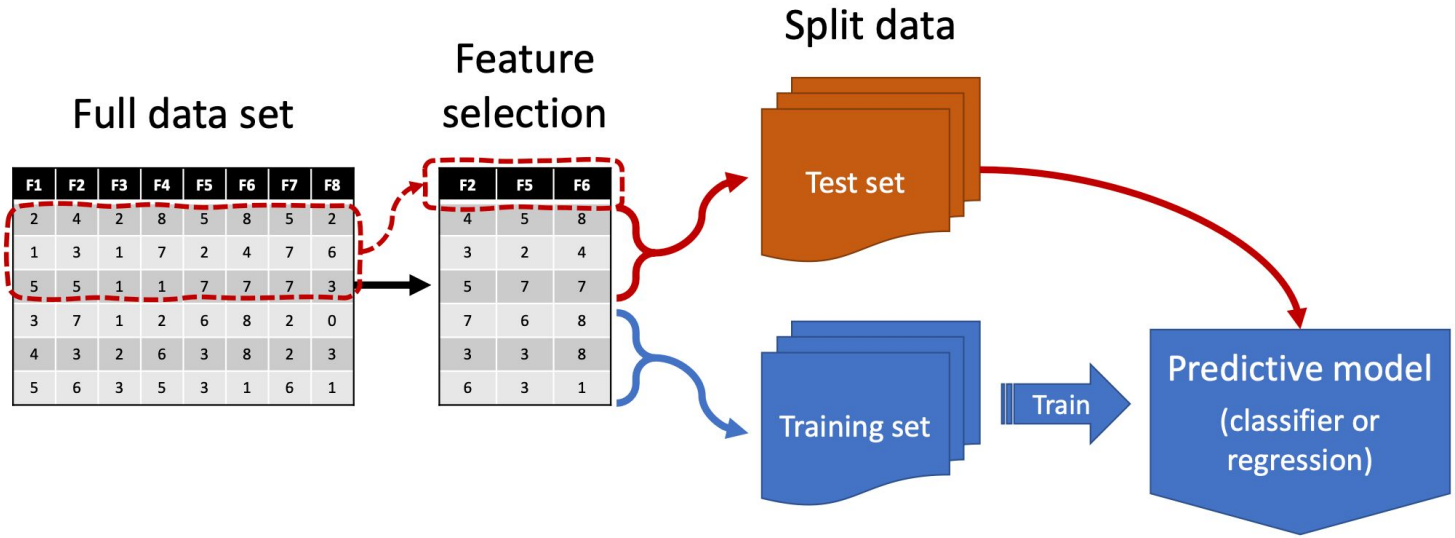




Lones (2024) *arXiv*

Common ML pitfalls

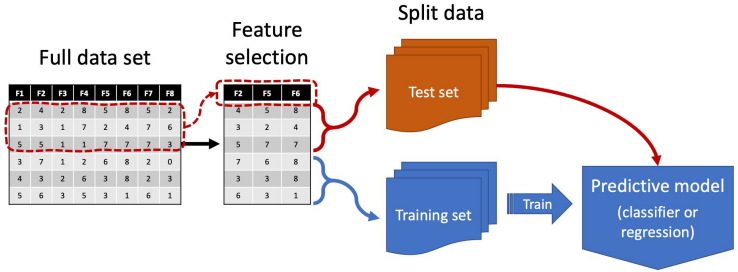
Data leakage

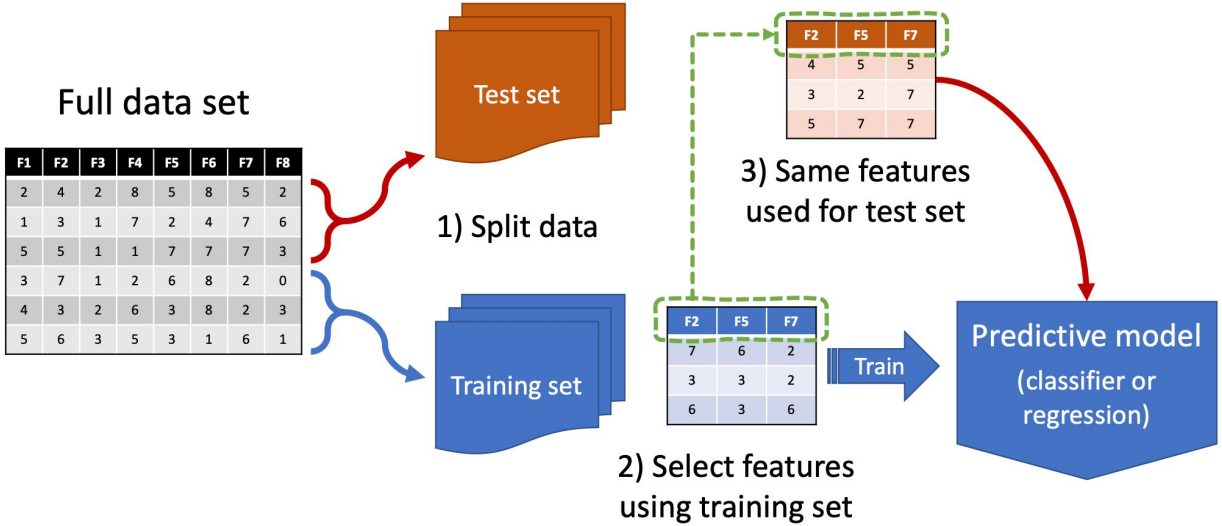


Lones (2024) *arXiv*

Common ML pitfalls

Data leakage





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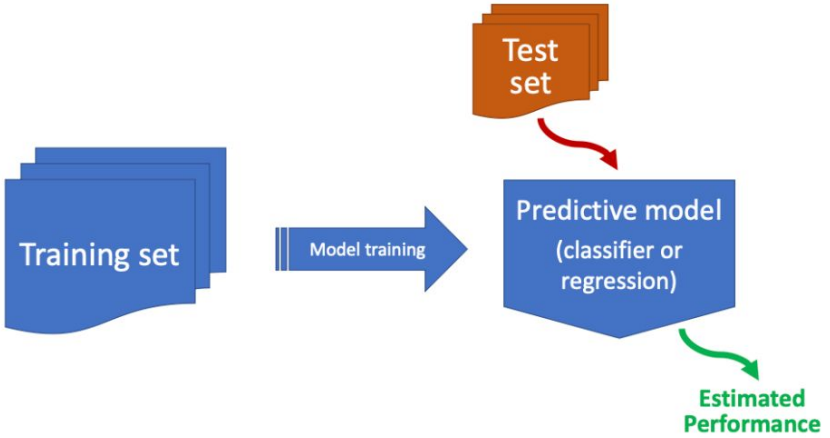
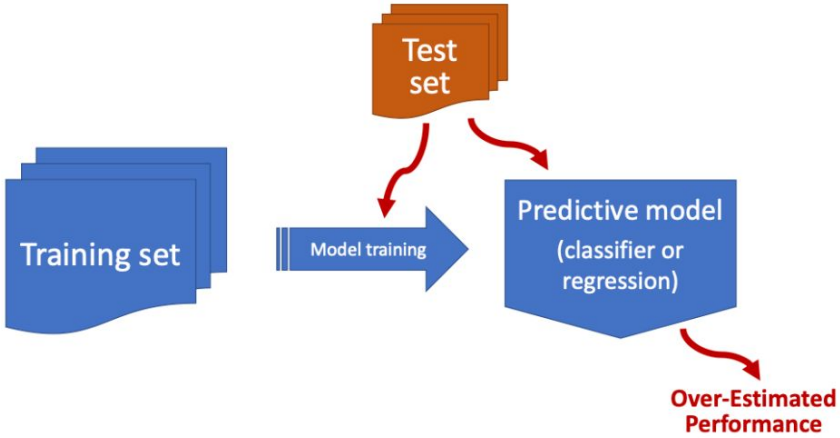
Common ML pitfalls

Data leakage





Exercise : connect this scenario to perform performing Principal Component Analysis (PCA) on a full dataset before splitting it into training and test folds. What's the issue & why?

Lones (2024) *arXiv*

Common ML pitfalls

Data leakage

Full data set

F1	F2	F3	F4	F5	F6	F7	F8
2	4	2	8	5	8	5	2
1	3	1	7	2	4	7	6
5	5	1	1	7	7	7	3
3	7	1	2	6	8	2	0
4	3	2	6	3	8	2	3
5	6	3	5	3	1	6	1
3	6	1	7	4	8	7	0
1	5	1	6	2	6	1	0
1	4	3	2	6	6	6	3
6	7	2	8	4	7	2	2
5	3	2	1	5	8	7	1
3	5	1	1	3	3	8	5
5	5	2	7	7	6	1	0
2	4	1	8	2	8	5	4
4	6	3	5	4	7	7	1

Cross-validation iteration 1

F2	F5	F6
7	6	8
3	3	8
6	3	1
6	4	8
5	2	6
4	6	6
7	4	7
3	5	8
5	3	3
5	7	6
4	2	8
6	4	7

F2	F5	F6
4	5	8
3	2	4
5	7	7

Cross-validation iteration 2

F2	F5	F6	F7
4	5	8	5
3	2	4	7
5	7	7	7
6	4	8	7
5	2	6	1
4	6	6	6
7	4	7	2
3	5	8	7
5	3	3	8
5	7	6	1
4	2	8	5
6	4	7	7

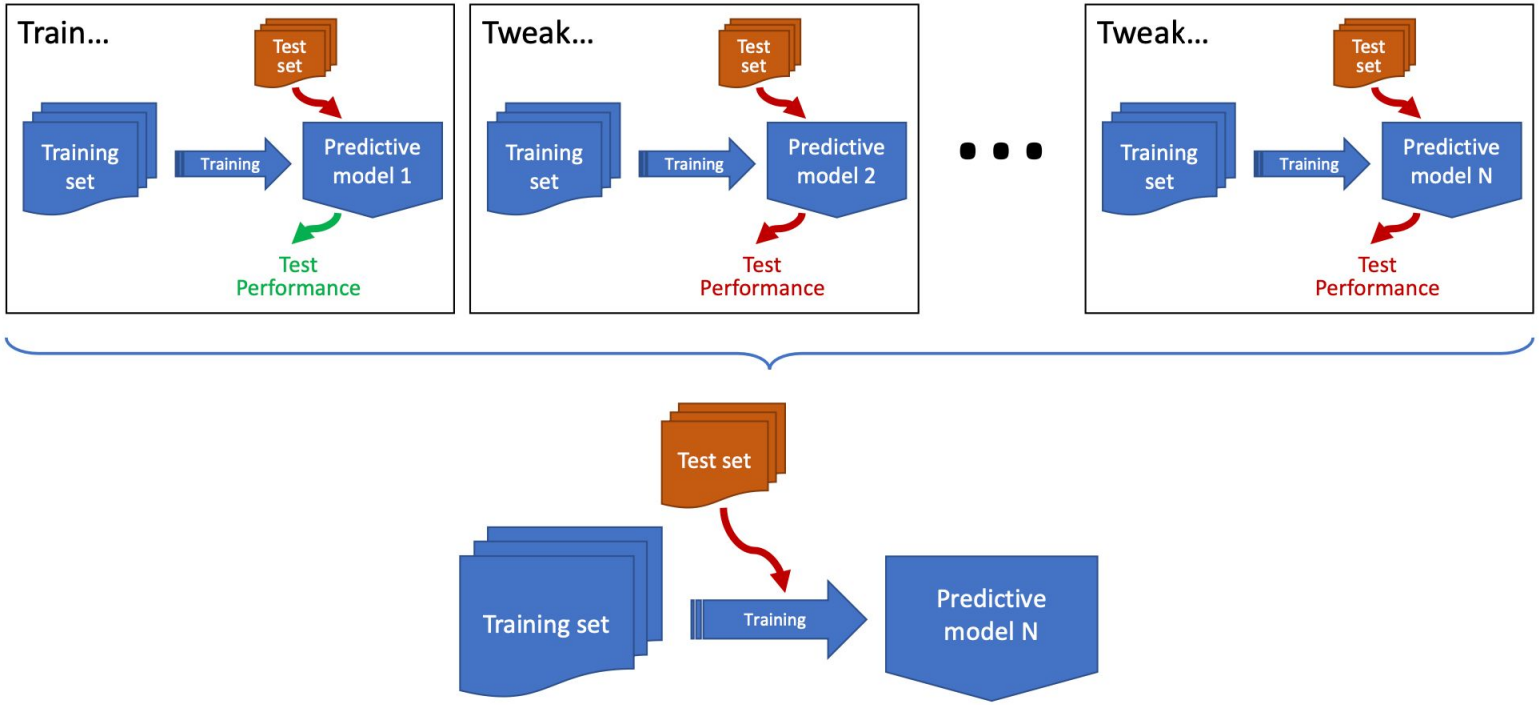
F2	F5	F6	F7
7	6	8	2
3	3	8	2
6	3	1	6

Independent feature selection for each iteration

Lones (2024) *arXiv*

Common ML pitfalls

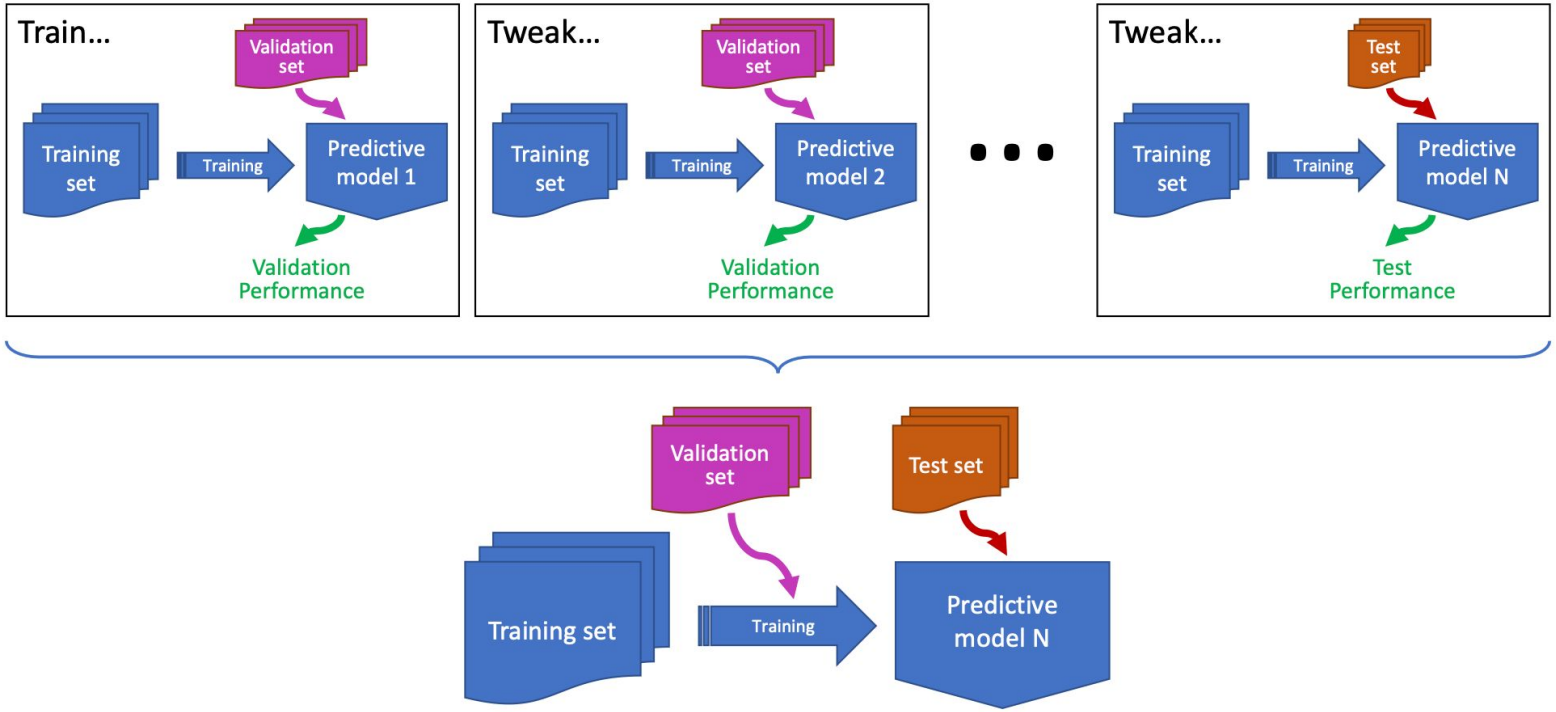
Sequential overfitting



Lones (2024) *arXiv*

Common ML pitfalls

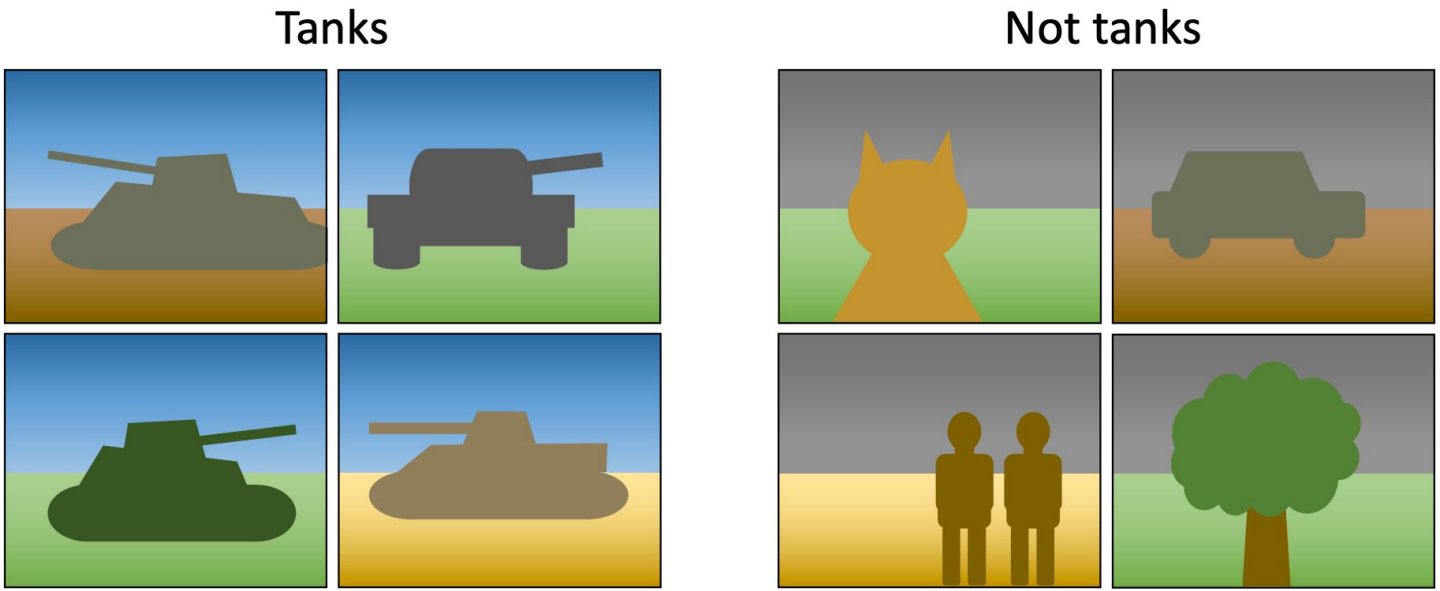
Sequential overfitting



Lones (2024) *arXiv*

Common ML pitfalls

Spurious associations



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Common ML pitfalls

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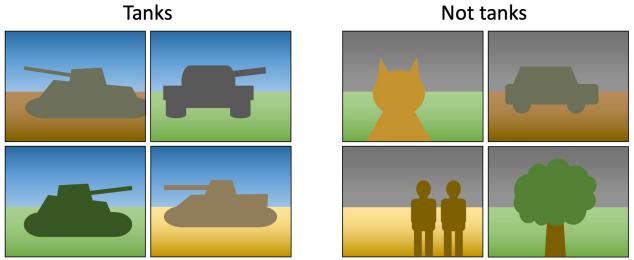
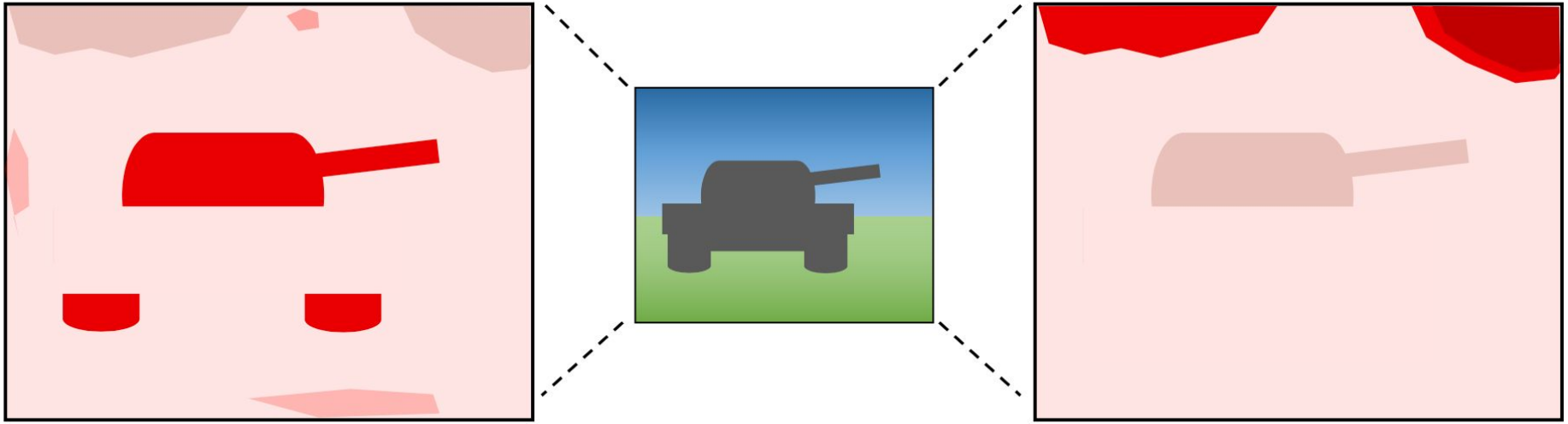
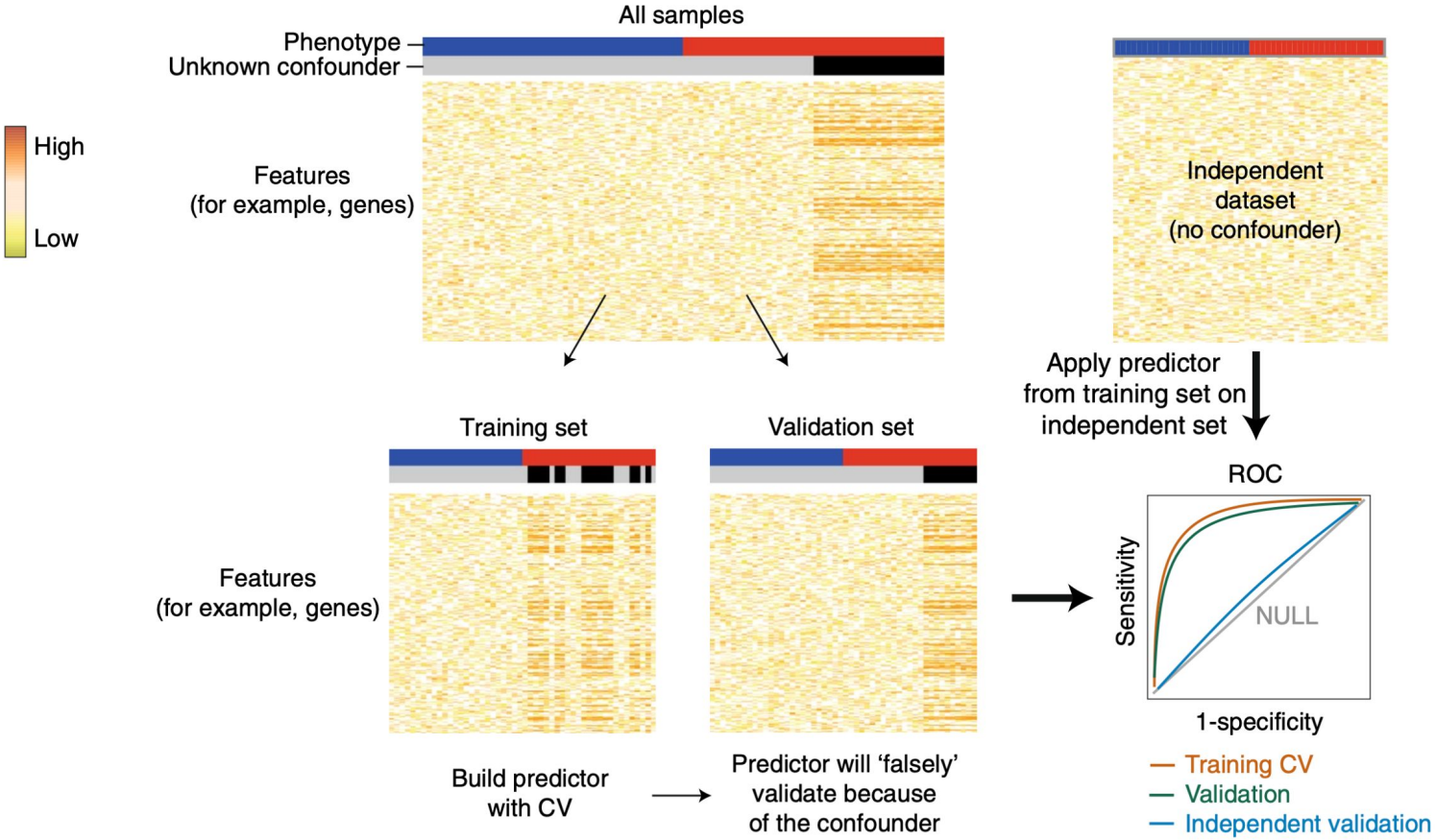




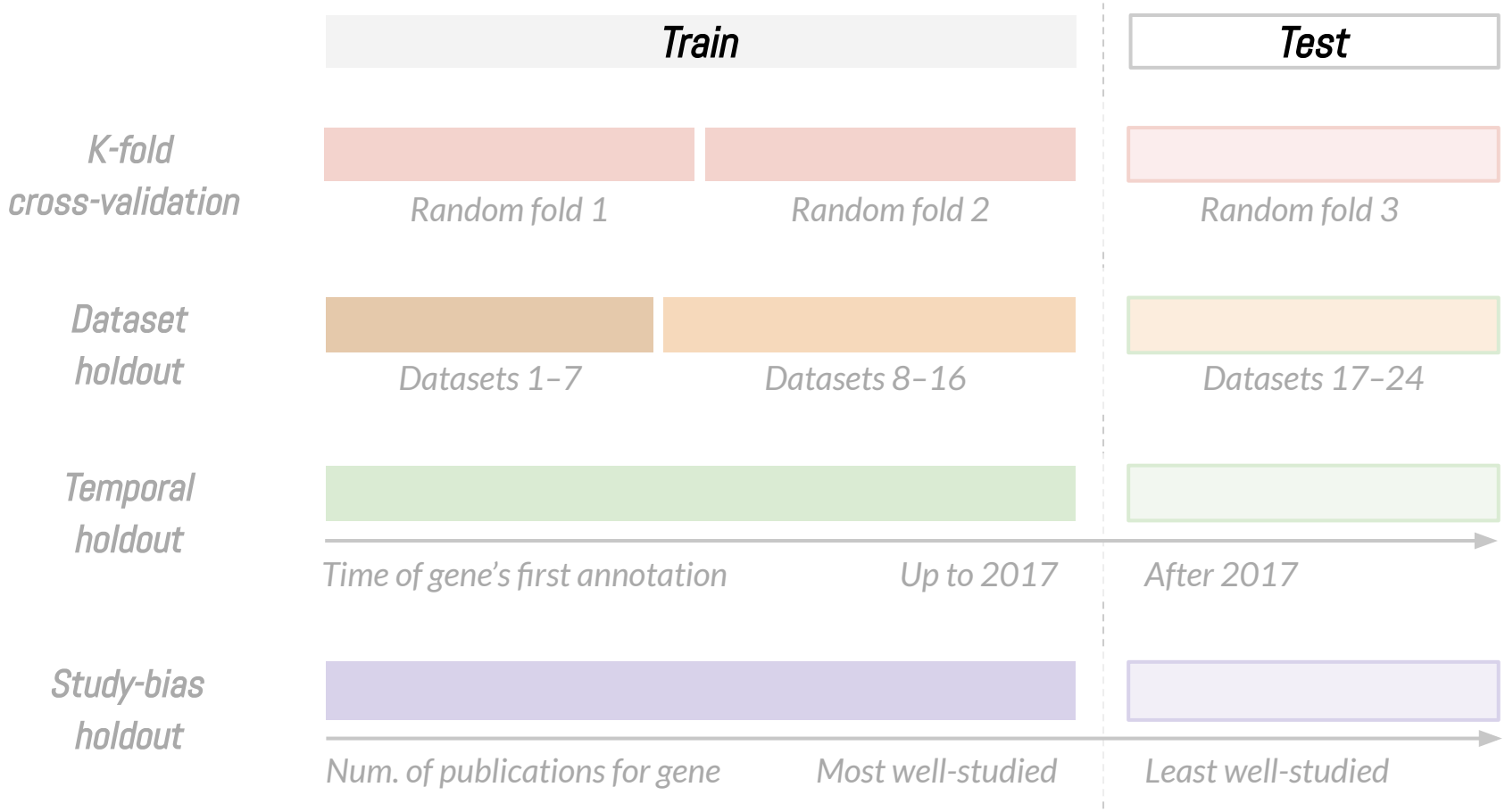
Lones (2024) *arXiv*

Common ML pitfalls



Teschendorff (2019) *Nat. Mat.*

Designing meaningful evaluation schemes



Common ML pitfalls

Inappropriate evaluation metrics

Nearly all biomedical prediction tasks are highly imbalanced:
negatives >> # positives
E.g., individuals with an adverse effect, genes in a pathway, or mutations linked to a disease

Popular metrics like the ***accuracy*** &
area under the ROC curve (auROC) inflate performance and are often misleading.

F1	F2	F3	F4	F5	Class
2	4	2	8	5	A
1	3	1	7	2	A
5	5	1	1	7	A
3	7	1	2	6	A
4	3	2	6	3	A
5	6	3	5	3	B
3	6	1	7	4	B
1	5	1	6	2	B
1	4	3	2	6	B
6	7	2	8	4	B

Always predict class A

Predicted	Correct?
A	TRUE
A	TRUE
A	TRUE
A	TRUE
A	TRUE
A	FALSE
A	FALSE
A	FALSE
A	FALSE
A	FALSE

$$\text{Accuracy} = \frac{5}{10} \quad \textbf{50\%}$$

Number of correct classifications

Total number of classifications

F1	F2	F3	F4	F5	Class
2	4	2	8	5	A
1	3	1	7	2	A
5	5	1	1	7	A
3	7	1	2	6	A
4	3	2	6	3	A
5	6	3	5	3	A
3	6	1	7	4	A
1	5	1	6	2	A
1	4	3	2	6	A
6	7	2	8	4	B

Always predict class A

Predicted	Correct?
A	TRUE
A	TRUE
A	TRUE
A	TRUE
A	TRUE
A	TRUE
A	TRUE
A	TRUE
A	TRUE
A	FALSE

$$\text{Accuracy} = \frac{9}{10} = \textbf{90\%}$$

Number of correct classifications

Total number of classifications

Lones (2024) *arXiv*

Common ML pitfalls

Inappropriate evaluation metrics

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negatives >> # positives
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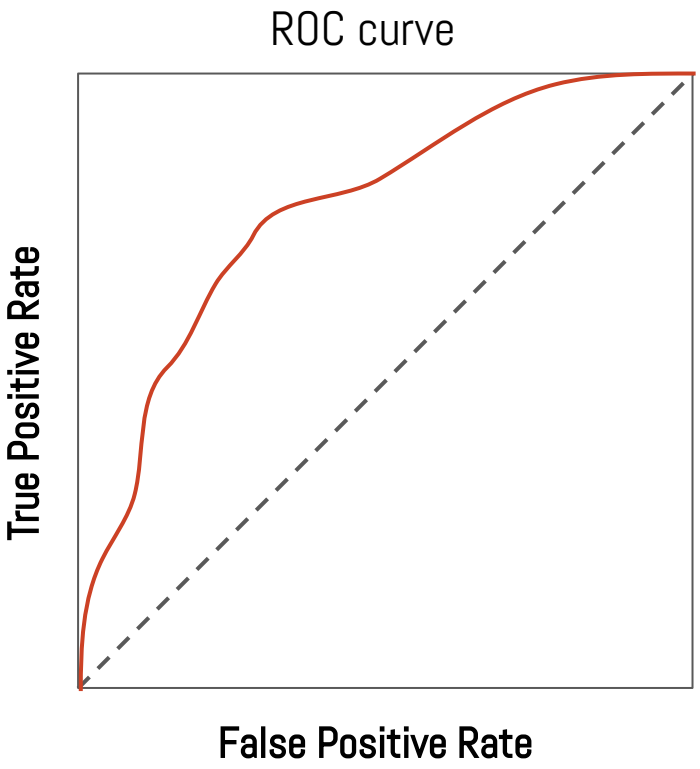

Common ML pitfalls

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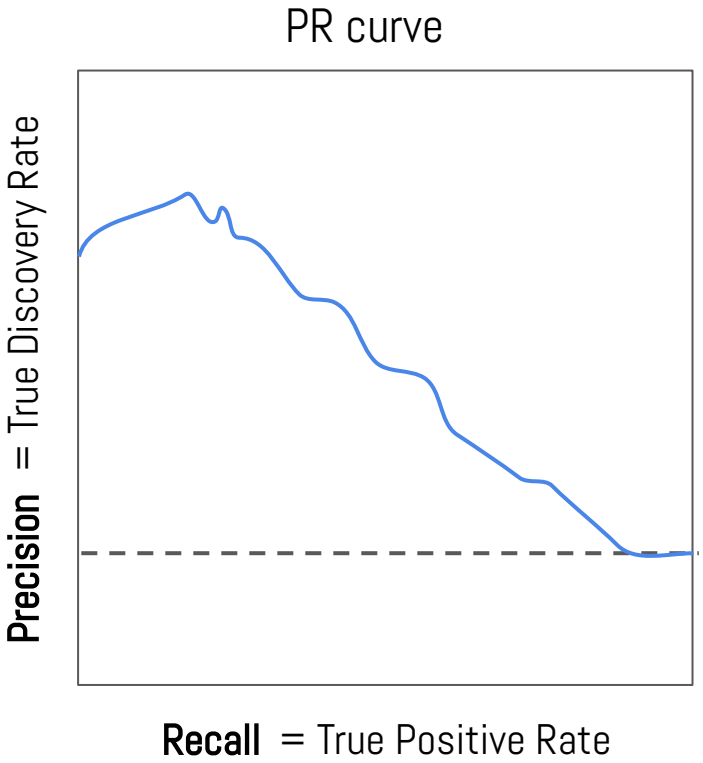
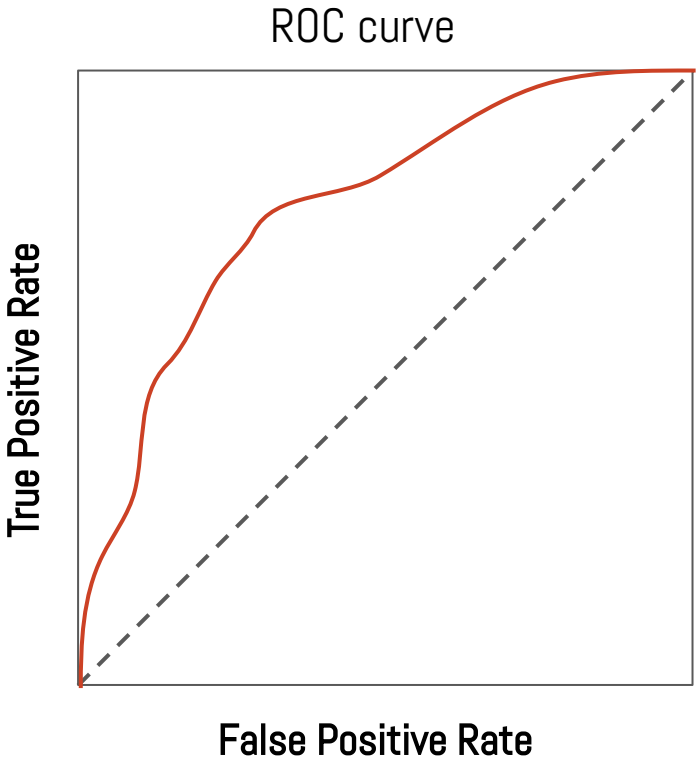
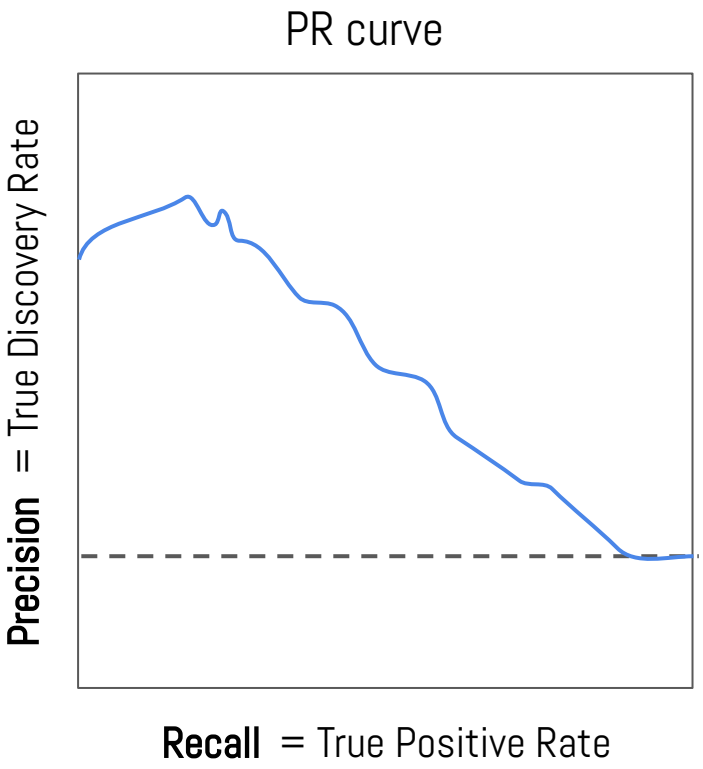
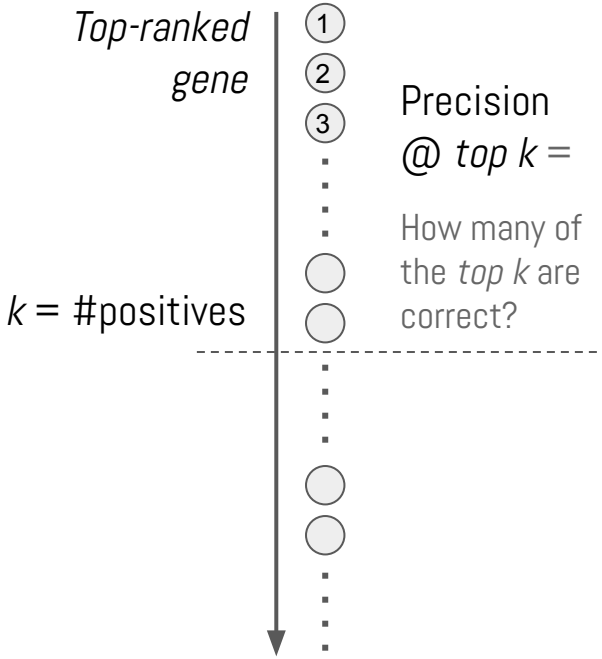




Choosing meaningful evaluation schemes

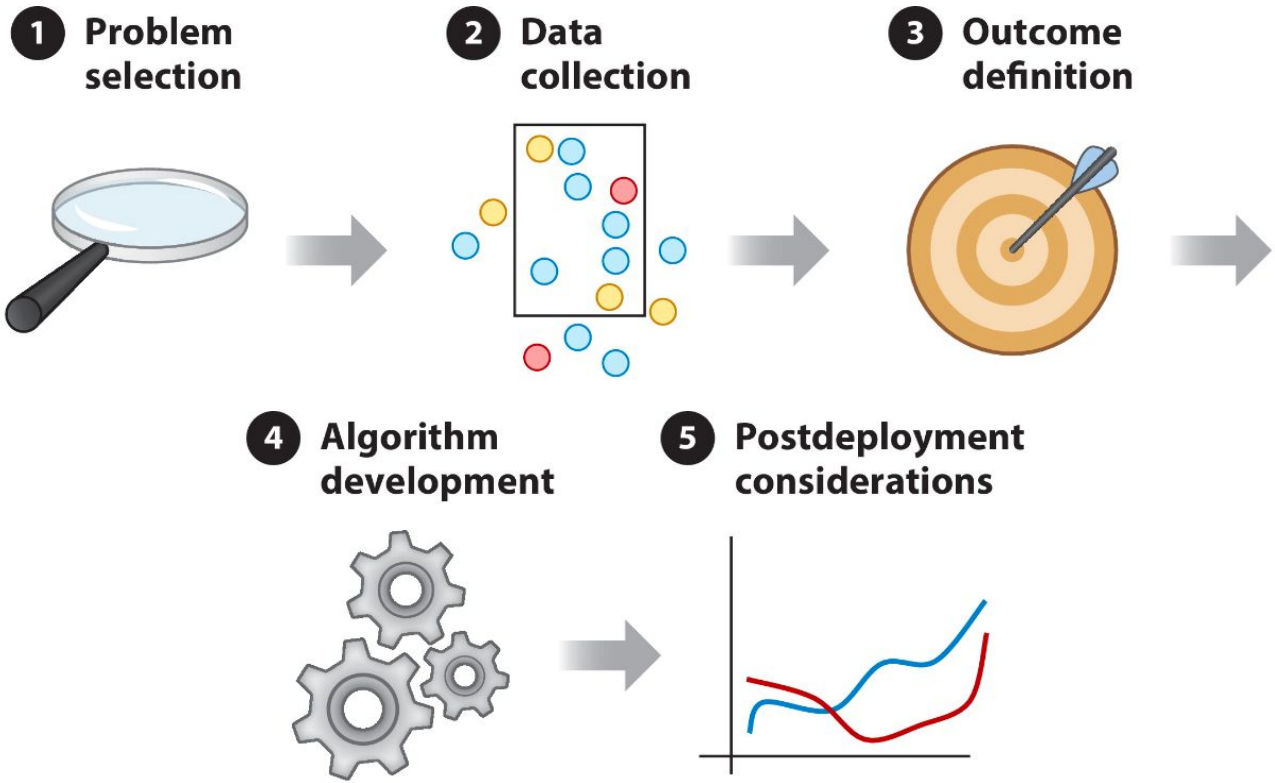




Model-free baselines

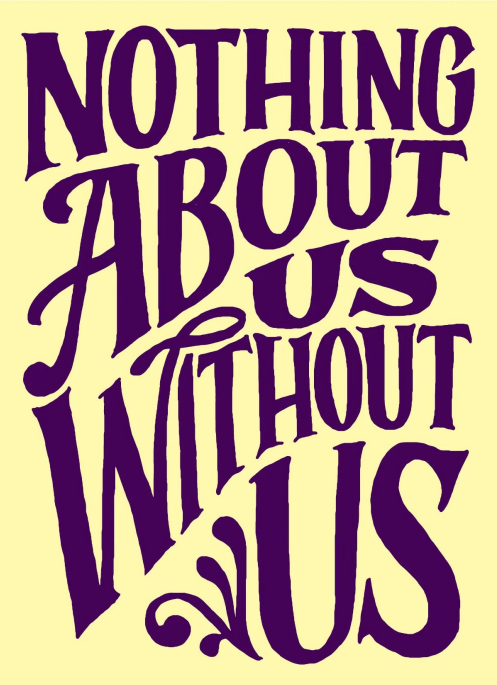
Real predictions	Length	Basal expression	Multifunctionality
gene 04	*gene 19*	*gene 32*	*gene 73*
gene 19	*gene 34*	*gene 26*	*gene 12*
gene 26	*gene 20*	*gene 40*	*gene 26*
gene 08	*gene 73*	*gene 67*	*gene 28*
gene 05	*gene 02*	*gene 16*	*gene 66*
gene 73	*gene 59*	*gene 19*	*gene 39*
gene 17	*gene 07*	*gene 05*	*gene 04*
...	...	...	...
...	...	...	...
...	...	...	...
...	...	...	...
...	...	...	...
...	...	...	...
...	...	...	...
...	...	...	...

Sources of bias, misalignment, and inequity in ML



Chen *et al.*, (2021) *Annu. Rev. Biomed. Data Sci.*

Who gets a seat at the table matters



Those affected by AI should get a voice in selecting what problems to address and guide the process.

Include lived experience. Get input from the most relevant participants.

E.g.: Contextual Analysis of Social Media

ML to help identify Twitter posts related to gang violence. Relies on the expertise of Black people who have experience with gangs in Chicago, Illinois. These experts review & correct notes underlying the algorithm.

Image: *ladyfingersletterpress.com*; Patton *et al.*, 2020 *Proc AAAI ACM Conf AI Ethics Soc.*

When you encounter an ML model, ask these Qs

1. **Benchmarking** :
   a. Test dataset's closeness to non-test/real data (incl. imbalance).
   b. Performance on out-of-distribution data points (i.e., data not identically distributed to training data).
   c. Performance on completely independent data (from different lab, hospital). Signs of data leakage / double dipping (between label-generation, training, & testing).
2. Choice of appropriate **performance metrics** . This includes accounting for properties like real-world imbalances and constraints of practical use (*e.g.*, want many predictions that could include false positives or want few sure predictions).
3. Comparison to **simple baselines** and state-of-the-art methods; Performance of the model on appropriately **randomized / perturbed data** .
4. Data and model properties associated with good/poor performance. Model's **failure modes** .
5. Statistical & **biomedical/practical significance** of improved performance (*e.g.*, the metric is higher, but is that going to translate to perceivable practical difference).
6. Requirements and constraints of **deploying the model in practice** (infrastructure, cost, use/misuse, etc.).

A sound computational analysis: controls, baselines, & robustness

Input → **ML model** → Output

What biological, statistical, and technical properties of the input data are associated with good or poor outputs?

Given the input and the desired output, what are simple methods that'll get the job done?

What is a scalable, high-throughput way to systematically evaluate / benchmark the outputs?

a. Evaluation datasets & groundtruth
b. Evaluation schemes
c. Evaluation metrics

What negative controls should be designed to assess whether the results are better than:

a. What can be trivially explained by global observational / statistical properties of the input data?
b. Data-free and model-free predictions?

What is the sensitivity of all the findings to changes in hyperparameters (incl. rand seeds) / model settings?